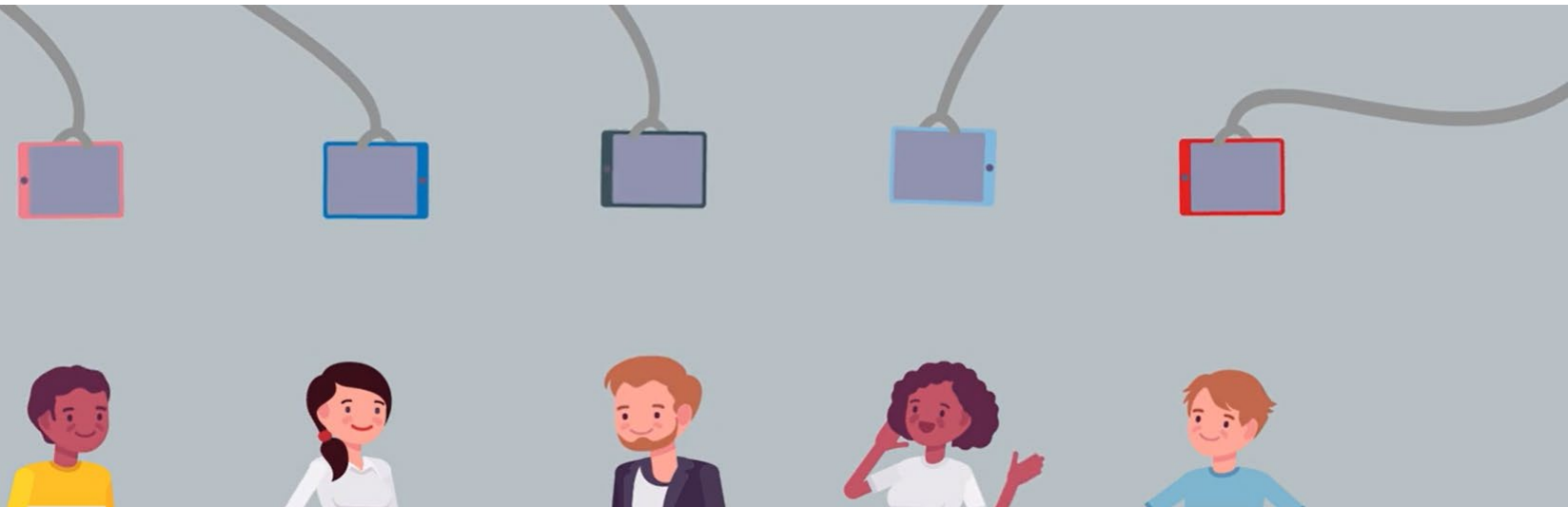


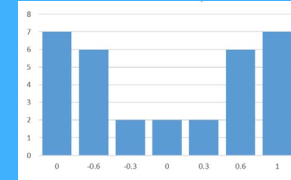
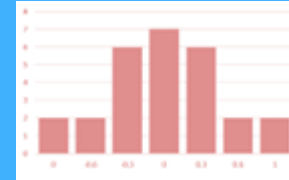


Juliane A. Lischka, Laura Laugwitz, Nadja Schaetz, Hendrik Meyer,
Lucien Heitz, Abdullah Rana, Jessica Kunert, Katharina Kleinen von Konigslow

When Recommender Systems Select Political News: Effects of Algorithmic News Curation on Party Preferences

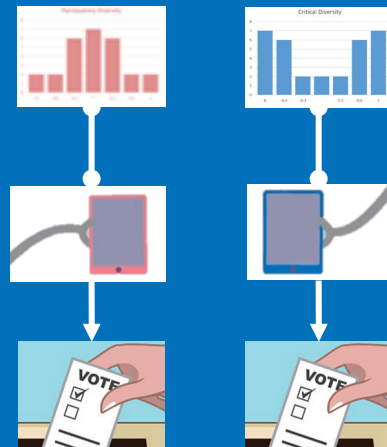
ICA 23, Toronto, CA; May 27, 2023

- NRSs can create distinct **party visibility patterns** (Helberger, 2019; Vrijenhoek et al., 2020)
- News **party visibility** might affect party preferences (Eberl, Boomgaarden, & Wagner, 2017; Geers & Bos, 2017; Hopmann et al., 2010)



Starting Points

*What are effects of varying party
visibilities in political news
recommendations on political
news usage, knowledge, attitudes,
and voting behavior?*



Research Question

Study Outline: Field Experiment

INPUT

OUTPUT

OUTCOME

WE
ARE
HERE

WE
ARE
HERE

Real-
world
news

→

Data-
base

→

Automated
content analysis

→

NRS
NRS

→

Control group
(n = 65)
Experimental group
(n = 78)

→

User
behavior

Political
knowledge,
attitudes,
voting behavior

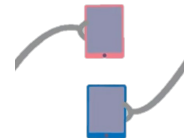


4,176 articles
from 6 outlets

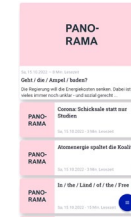
centerShip to site indexPublicSubscribLog InToday's
Who Criticized Trump: PERSON in Texts, Is Freedom of the Press
Investigation after his disparaging tweets about President Trump PERSON were uncovered, was fired
Tenshly Adam Goldman OBC and Michael S. SchmidtAug PERSON 13 CARDINAL 2018WA
PERSON the F.B.I. OPE senior counterintelligence agent who disparaged President Trump PERC
oversees the Hillary Clinton PERSON email and Russia OPE investigations, has been fired for viola
sat Monday DATE Mr. Trump and his allies seized on the tweets — exchanged during the 2016 SA
Lisa Page — in PERSON assailing the Russia OPE investigation as an illegitimate "witch hunt." M

2,134 articles with
party reference

News as
published
Priority for
small-party
news

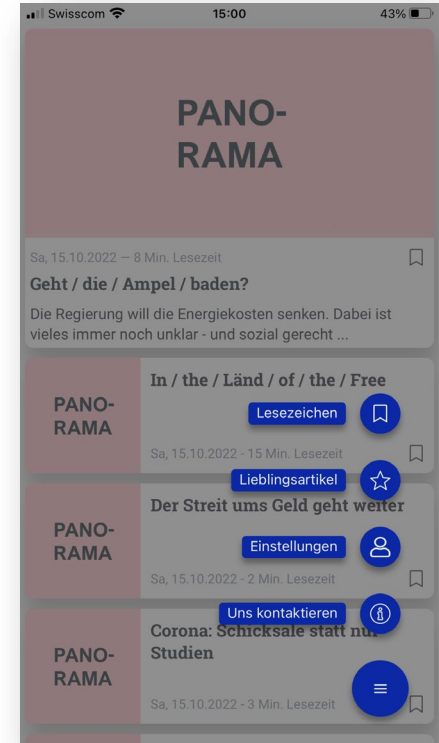
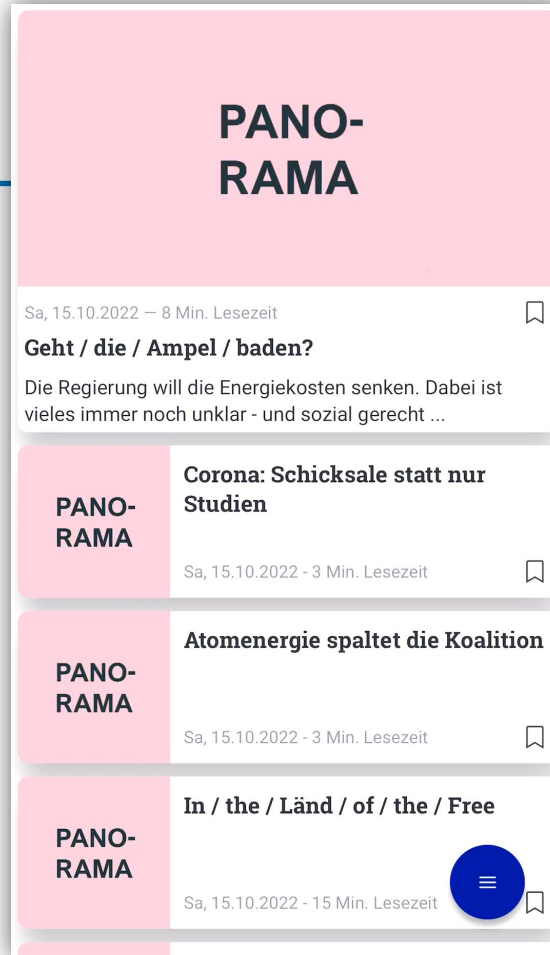


Ex-ante
online
survey

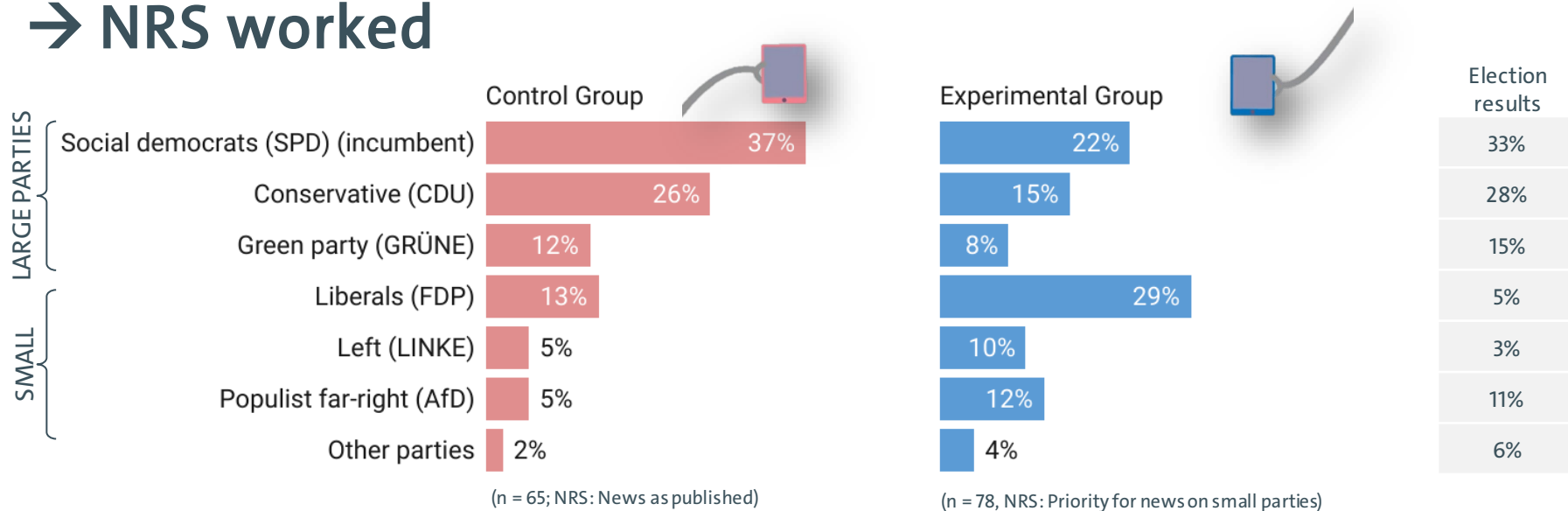


Ex-post
online
survey

News App Interface

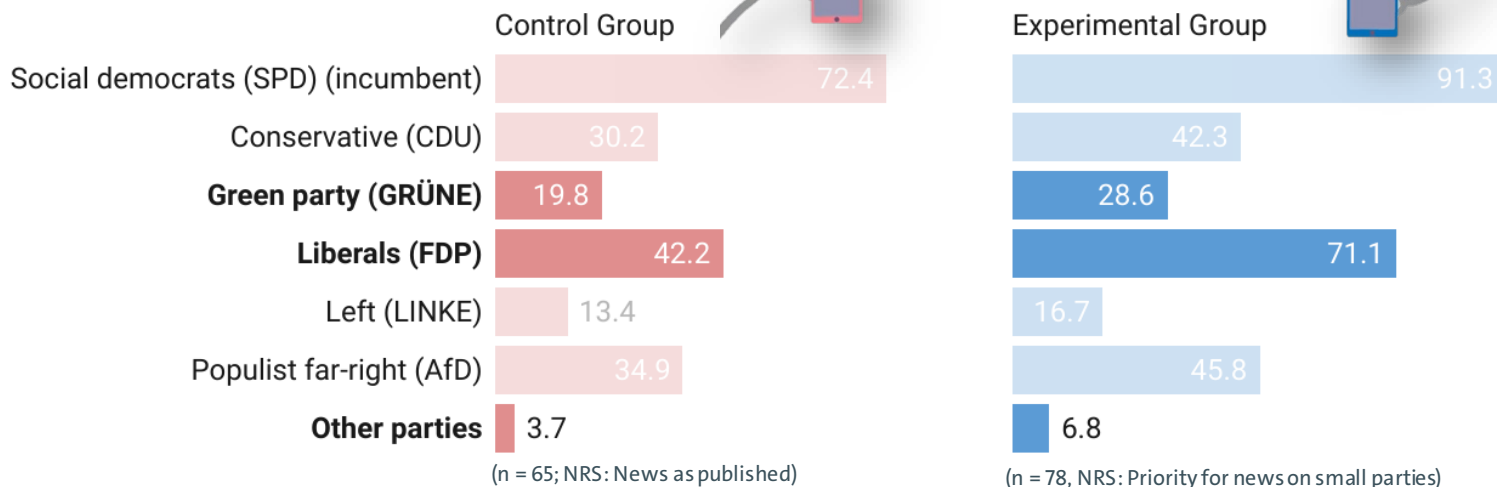


Supply: All party visibilities differ significantly → NRS worked



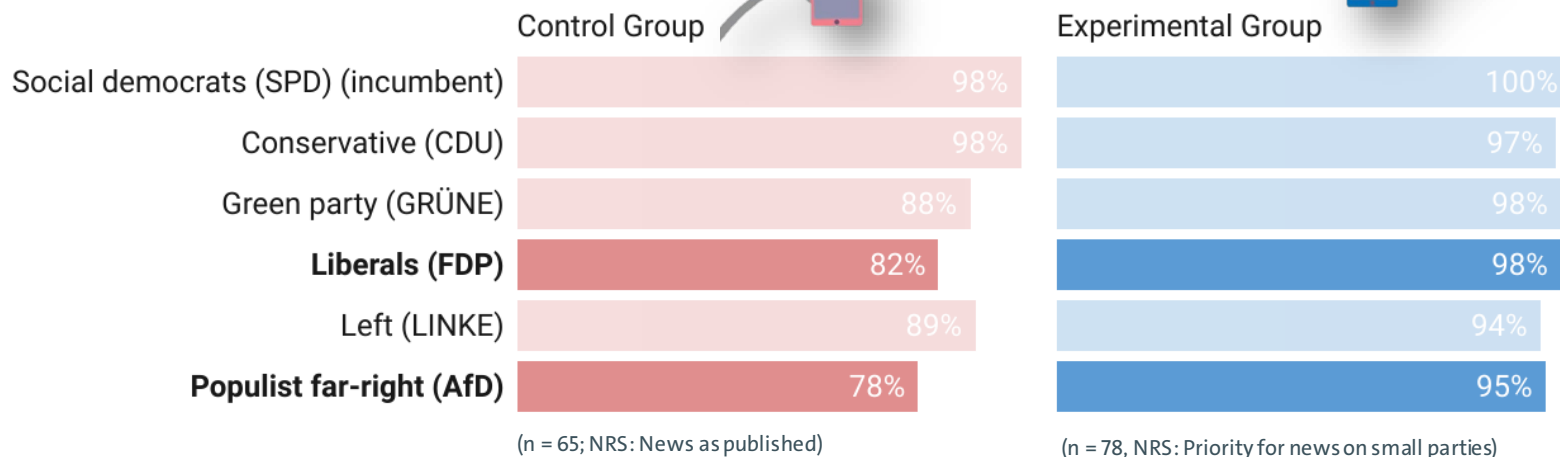
Created with Datawrapper

Demand: Higher usage intensity in experimental group for Green, Liberal, Others



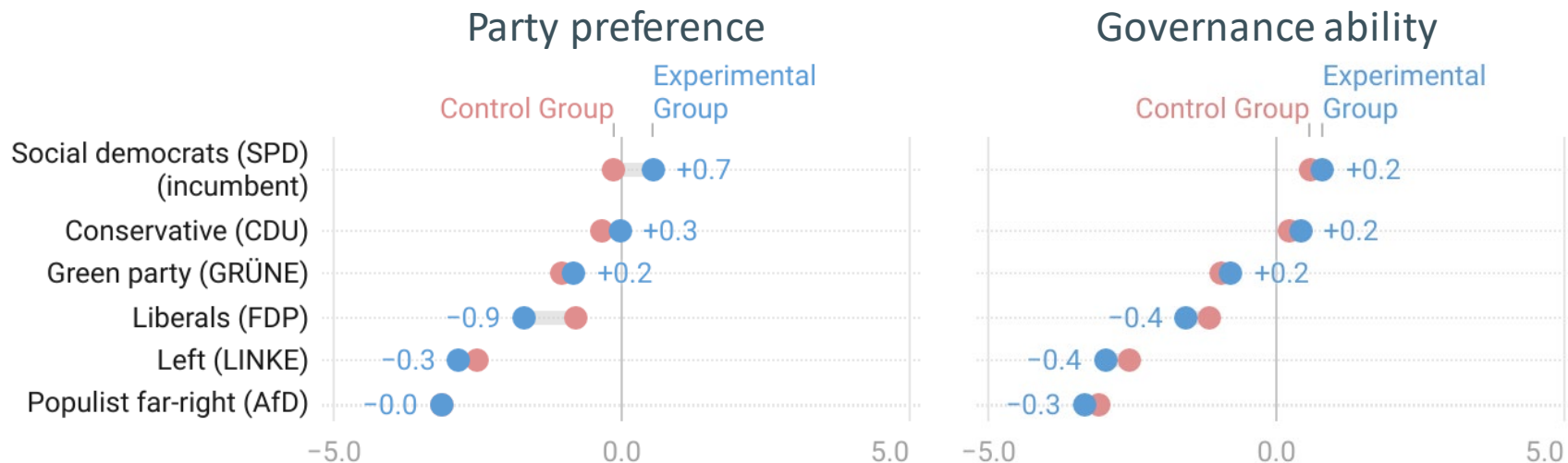
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Cognition: Knowledge (correct candidate attribution) differs for Liberal and Far-Right



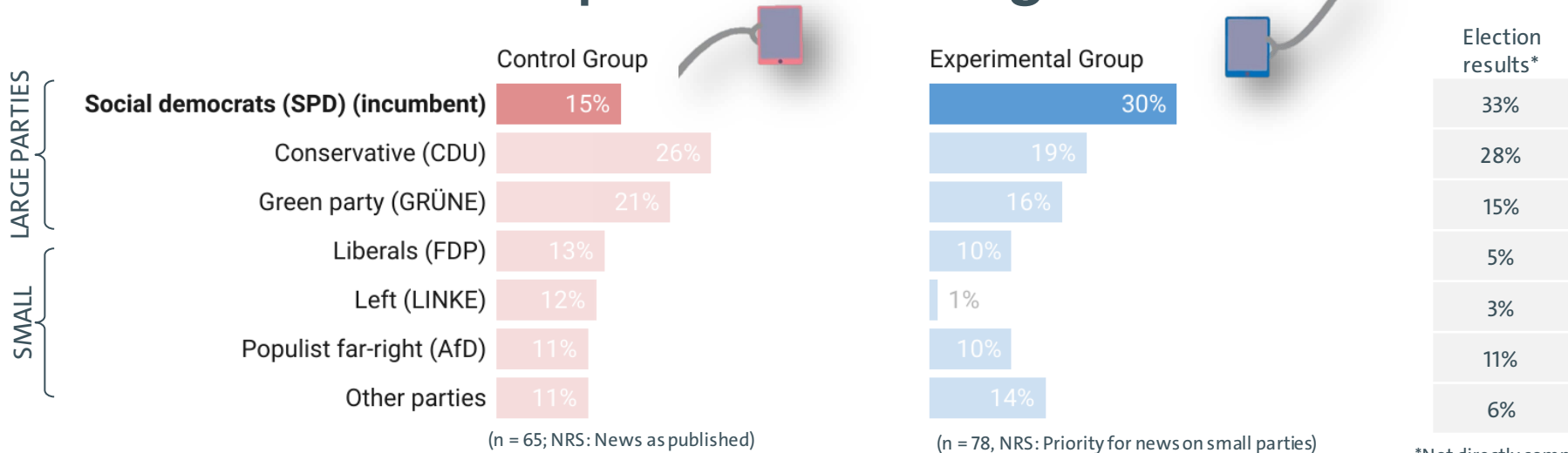
Created with Datawrapper

No significant attitudinal differences



Created with Datawrapper

Voting behavior: Difference for Social Democrats, BUT unrelated to experimental design



Created with Datawrapper

*Not directly comparable as we over-recruited small-party supporters.

Discussion

- USAGE: exposure to more small-party news is partly associated with higher reading intensity
 - more informed voting decision
- KNOWLEDGE: exposure to more small-party news is partly associated with an increase in knowledge about their candidates
 - more informed voting decision
- No effects on ATTITUDES
- No effects on VOTING BEHAVIOR
 - No translating to a change in voting behavior

CONCLUSION

Even though effects on usage & knowledge level only; we should invest into **socially responsible** (= content-diverse) NRS!

THANK YOU!

Ethical Measures

1. Transparent goal, procedure, and risk explanation to participants before participation agreement
2. No party should have a visibility advantage or disadvantage so that any effects even out overall participants

		Control condition (news as published) (n = 65)					Experimental condition (higher small-party visibility) (n = 78)						Election results	
		OUTPUT	OUTCOME				OUTPUT	OUTCOME						
		Supply Party visibility	Demand Usage intensity	Cognition Knowledge	Attitudes Party preference	Behavior Governan- ce ability	Voting decision	Supply Party visibility	Demand Usage intensity	Cognition Knowledge	Attitudes Party preference	Behavior Governan- ce ability		Voting decision
LARGE PARTIES	Social demo- crats (SPD) (incumbent)	37% ^a	72,42	98%	-.12	.59	15% ^a	22% ^b	91,31	100%	.55	.81	30% ^b	33%
	Conservative (CDU)	26% ^a	30,18	98%	-.34	.22	26%	15% ^b	42,26	97%	-.03	.42	19%	28%
	Green party (GRUNE)	12% ^a	19,77 ^a	88%	-1.02	-.95	21%	8% ^b	28,61 ^b	98%	-.82	-.78	16%	15%
SMALL PARTIES	Liberals (FDP)	13% ^a	42,17 ^a	82% ^a	-.77	-1.16	13%	29% ^b	71,06 ^b	98% ^b	-1.68	-1.55	10%	5%
	Left (LINKE)	5% ^a	13,37	89%	-2.48	-2.53	12%	10% ^b	16,69	94%	-2.81	-2.96	1%	3%
	Populist far- right (AfD)	5% ^a	34,90	78% ^a	-3.11	-3.06	11%	12% ^b	45,75	95% ^b	-3.12	-3.32	10%	11%
	Other parties	2% ^a	3,65 ^a		-.60	-1.00	11%	4% ^b	6,83 ^b		1.60	4.00	14%	6%

Usage intensity

„Political Consumption Score“

Combination of user metrics regarding ...

- „Party Reference Score“ per article
- Reading time per article
- Scrolling activity per article



Content analysis

Sentiment

- Pre-trained ML
German sentiment Transformer
- So far:
just Sentiment, no differentiation
between emotions
- Best practice on
comparability between docs?

2	negative	Seit Beginn des Kriegs gegen die Ukraine hat R...
3	neutral	Gefolgt sind sie ihm trotzdem - und zu Beginn ...
4	neutral	"Auch wir sind einem Stresstest unterworfen" ...
5	neutral	Doch wenn eine Antwort gefunden sei, stünden d...
6	positive	"Nie habe ich mich so zuhause gefühlt wie in d...
7	neutral	Der Partei geht es umgekehrt offenbar genauso.
8	neutral	Immer wieder gibt es langen Applaus während se...
9	neutral	Es lohnt sich, an der Regierung zu sein, das w...
10	neutral	Auch wenn das Kompromisse bedeutet - etwa beim...
11	neutral	Das für Habeck und Partei heikelste Thema alle...
12	neutral	Dann sollte das Delegiertentreffen debattieren...
13	neutral	Dezember hinaus laufen zu lassen.
14	neutral	Als "Einsatzreserve" zur Netzstabilisierung so...
15	neutral	Das allein ist für große Teile der Basis, die ...
16	neutral	Doch Habeck könnte davon profitieren, dass ein...
17	neutral	Der will nicht nur, dass alle drei noch aktive...
18	neutral	Ob die Partei sich hinter Habeck stellt, stand...
19	neutral	Die Partei rechnete mit Debatten bis in den sp...

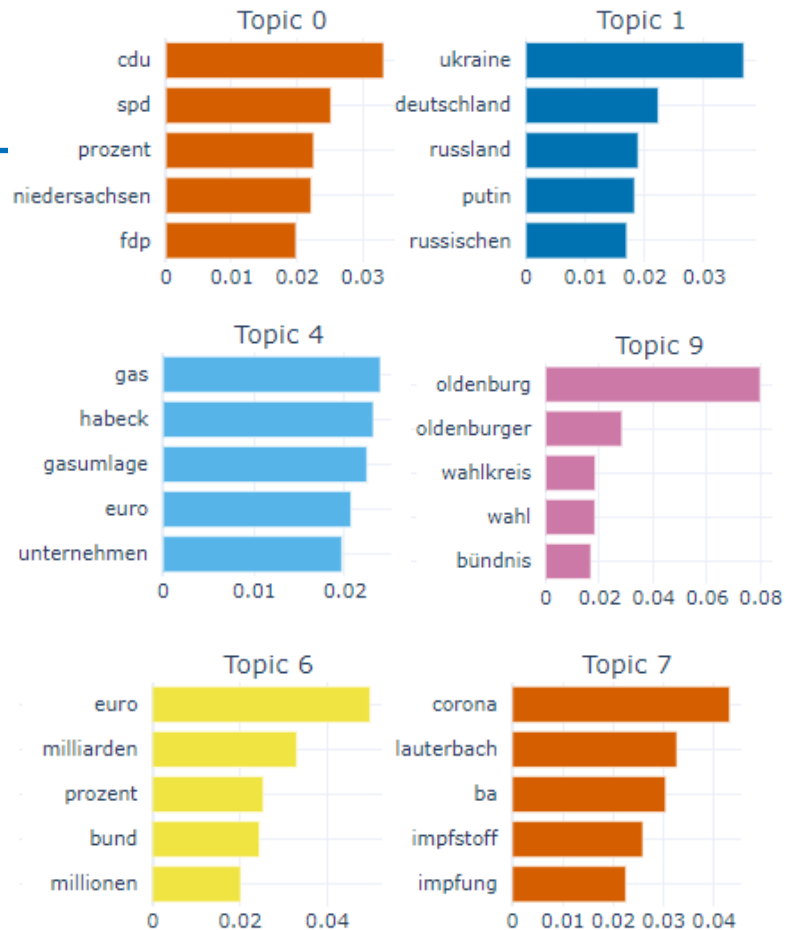

```
[ ] df.output.value_counts().to_frame('count').join(df.output.value_counts(normalize=True).to_frame('%'))
```

	count	%
neutral	17	0.85
negative	2	0.10
positive	1	0.05

Content analysis

Agenda / topics

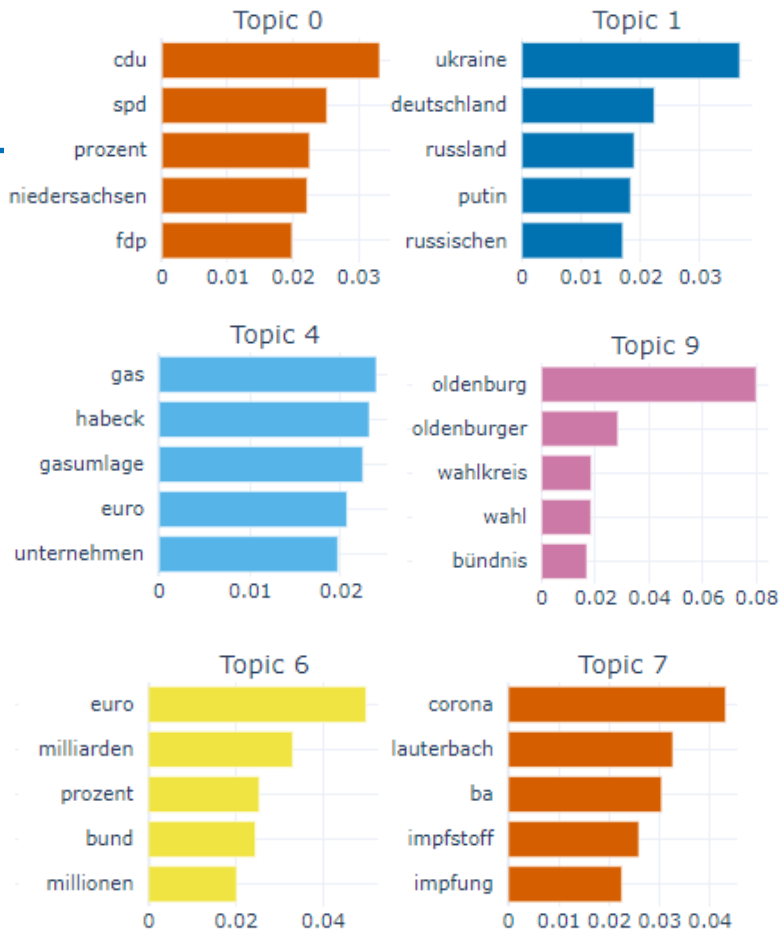
- BERTopic
(Transformer + UMAP + Clustering + c-TF-IDF)
- Provides a great overview



Content analysis

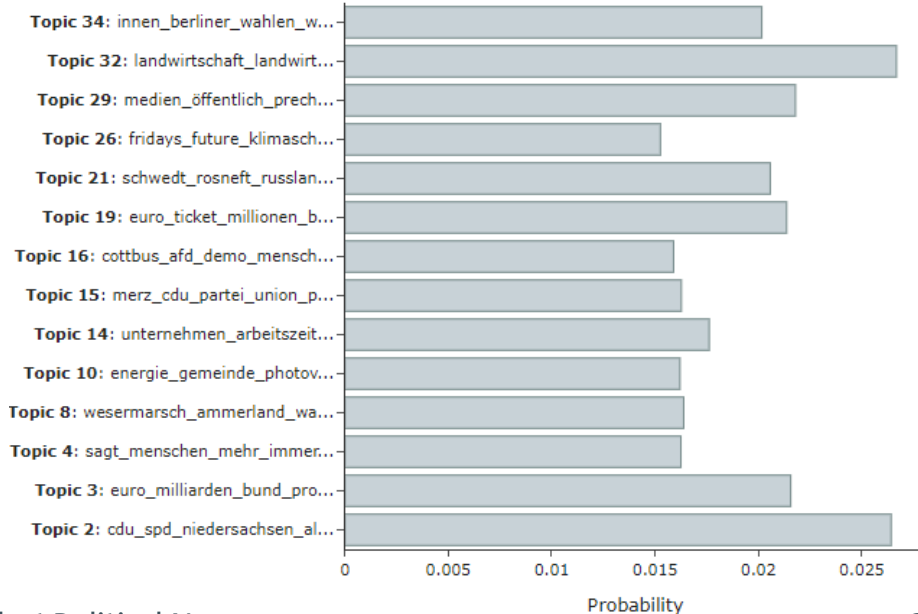
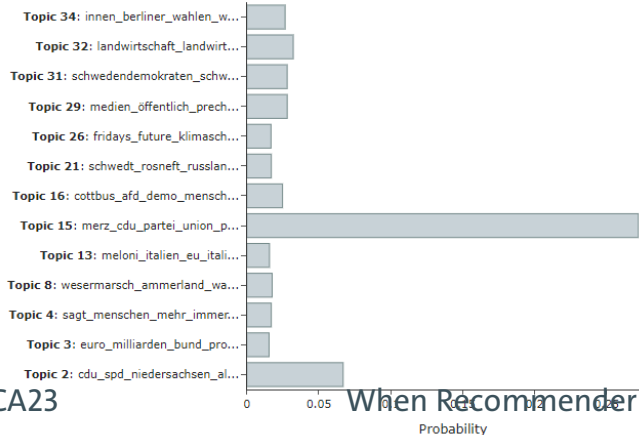
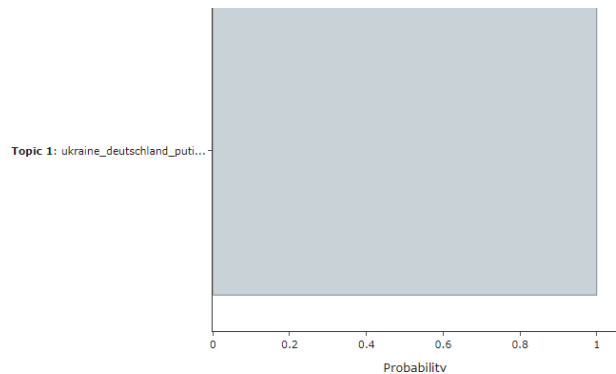
Agenda / topics

- BERTopic
(Transformer + UMAP + Clustering + c-TF-IDF)
- Provides a great overview
- But:
Assigns one topic per text/paragraph
→ diverse texts == outliers



Content analysis:

Topic Probability Distribution per Document



Next Steps?

Combining insights:

Conditions connected to ...

- Topics + Party (Dictionary + sentences/paragraphs?)
- Sentiment + Party (Dictionary + sentences/paragraphs?)
- Consumption

→ Participants' change in attitudes?

Content analysis

- Party visibility (PRE-REC)
- Agenda / topics (POST-REC)
- Sentiment (POST-REC)

User behavior

- Political Consumption
- Topical context
- Sentiment context

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said Monday **DATE** .Mr. Trump and his allies seized on the texts —

Lisa Page — in **PERSON** assailing the **Russia GPE** investigati



Content analysis

- Party visibility (PRE-REC)
- **Agenda / topics (POST-REC)**
- **Sentiment (POST-REC)**

User behavior

- Political Consumption
- **Topical context**
- **Sentiment context**

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Content analysis

Party visibility (PRE-REC)

- Dictionary-based
- „Political Reference Score“ per text

→ sampling and recommendation:

4176 articles,
2134 incl. political reference

Condition 1
(news as published)



Condition 2
(higher small-party visibility)

